



Do recommended doses of glyphosate-based herbicides affect soil invertebrates? Field and laboratory screening tests to risk assessment

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HIGHLIGHTS

- Recommended doses of glyphosate herbicides were applied to desiccate oat.
- Bait lamina (field) and laboratory tests were run to screening the ecotoxicity.
- Earthworms and isopods not avoided soil and oat straw from field in multispecies test.
- Reproduction of collembolans was not impaired but they avoided one product.
- Reduction on soil fauna feeding activity was observed to the red label product.

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ABSTRACT

Despite glyphosate-based herbicides are widely used in agriculture, forestry and gardens, little is known about its effects on non-target organisms. The present work evaluated the ecotoxicity of four formulated products (Roundup® Original, Trop®, Zapp® Qi 620 and Crucial®) on soil invertebrates. Screening ecotoxicity tests were carried out with soil and oat straw collected in a field experiment, besides laboratory-spiked soils. Screening tests included avoidance behaviour of earthworms (*Eisenia andrei*), collembolans (*Folsomia candida*) and isopods (*Porcellio dilatatus*) in single and multispecies tests; reproduction of collembolans (*F. candida*), and bait lamina in field. Non-avoidance behaviour was observed in standard tests (earthworms) in soil, neither in multispecies tests (earthworm + isopods) using oat straw, while for collembolans it occurred for the product Zapp® Qi 620 even at the recommended dose. Reproduction of *F. candida* was not impaired even at high doses in laboratory-spiked soils. Feeding activity on bait lamina test was impaired in treatment corresponding to the red label product, Crucial®. Results showed the relevance of bait lamina test on screening the impact of herbicides in the field. The findings highlight the importance of considering different formulations for the same active ingredient in risk assessment of pesticides.

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1. Introduction

Nowadays, agricultural intensification is heavily dependent on herbicides for weed control (Allegri et al., 2015). Considered the major agrochemical consumer's country (Carneiro, 2015), the main

reasons for increasing consumption of herbicides in Brazil are agriculture frontiers' expansion and increasing on land use for no-till practice (Tavella et al., 2011).

As the most popular herbicide used worldwide, the acid equivalent glyphosate [(N-phosphonomethyl) glycine] will probably remain as the world's most used agrochemical during the next years (USEPA, 2012) and thus the interest in quantifying its impacts on the environment and human health are needed (Benbrook, 2016). It is a non-selective systemic broad-spectrum inhibitor of the enzyme 5-enolpyruvylshikimate-3-phosphate synthase (EPSPs) which is not present in human or animal cells. EPSPs

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inhibition causes the interruption of three aromatic amino acids (phenylalanine, tyrosine and tryptophan) vital to plant synthesis (Giesy et al., 2000; Lane et al., 2012). Although the foliar application, a great amount of glyphosate can reach the soil (Lane et al., 2012) causing adsorption by minerals and organic phase (Maqueda et al., 2017) leading the residue to soil invertebrates.

Despite glyphosate-based herbicides are widely used in agriculture, forestry and gardens, surprisingly little is known about their effects on non-target organisms (Zaller et al., 2014). The existent studies on glyphosate ecotoxicity and its metabolites for soil fauna are still non-conclusive (Domínguez et al., 2016). While Santos et al. (2012) found no effects for collembolans and earthworms in glyphosate recommended doses applied in Mediterranean soil, Casabé et al. (2007) related damage to soil fauna and ecosystem services in areas with glyphosate application in soybean plantation in Argentina. Kanashiro (2015) found chronic effects for earthworms just in high doses, otherwise another study involving one of the main metabolites of glyphosate, aminomethylphosphonic acid (AMPA) pointed out that earthworms from parents exposed to contaminated soil can have size and growth restricted and may affect ecosystem function carried out by them in agricultural or forestry soil (Domínguez et al., 2016).

Other uncertainties concerns to the herbicides formulation. Commercially formulated as potassium, isopropyl amine, monoammonium, diammonium and trimesium salts, these formulations may include inert substances known as surfactants which, in some cases, can have higher toxicity when compared to the acid equivalent itself (Giesy et al., 2000; Aguiar et al., 2016).

The aim of this work was to evaluate four commercial glyphosate herbicides, applied in recommended doses in field and higher doses in laboratory, presenting a screening of its ecotoxicity and expected effects on soil invertebrates.

2. Material and methods

2.1. Field experiment

Four commercial glyphosate herbicides were used in this study:

- Roundup® Original: 360.00 g L⁻¹ acid equivalent (a.e.) glyphosate, isopropylamine salt – 480.00 g L⁻¹, soluble (liquid) concentrate - SL;
- Trop®: 354.61 g L⁻¹ a.e. glyphosate, isopropylamine salt – 480.00 g L⁻¹, SL;
- Zapp® QI 620: 500.00 g L⁻¹ a.e. glyphosate, potassium salt 620.00 g L⁻¹, SL;
- Crucial®: 540.00 g L⁻¹ a.e. glyphosate, isopropylamine salt 400.8 g L⁻¹ and potassium salt 297.75 g L⁻¹, SL.

These herbicides were applied in field to burndown black oat in recommended dose of 720 g L⁻¹ a.e. for each product, evaluated at the Experimental Farm of Federal University of Santa Catarina (UFSC), in Curitiba, SC, Brazil, in three plots per treatment, in a randomized block design. Application using a pressurized sprayer with CO₂ was used with a 2.0 m long bar containing four sprays tips AVISO 11002 interspaced 50 cm from each other at a distance of 50 cm of the cultivation, kept at a work pressure of 2.8 kgf cm⁻² and speed of 3.6 km h⁻¹ providing an application volume of 200 L ha⁻¹. Three plots without application of herbicides were used as control. Soil was classified as haplic cambisol (2.2% organic matter, clay texture) and climate is classified as Cfb, with mean temperatures reaching less than 22 °C in the warmer month and annual precipitation around 1400 mm.

2.2. Collection and preparation of the samples for ecotoxicity tests

Eleven days after the field application, soil samples were collected at surface layer (20 cm), sieved (5 mm mesh) and immediately used in ecotoxicity tests. The moisture was adjusted to 50% of the water holding capacity and pH measured at the beginning and at the end of each test.

Oat straw samples were collected at each treatment, transported to the lab and cut in 2 cm size fragments using a paper-cutter. This process was adopted in order to facilitate the use of oat straw in avoidance tests, standardizing the volume of straw in each side of test's recipients.

2.3. Preparation of contaminated soil in laboratory

Additional tests were carried out with lab -contaminated soil with higher concentrations than that expected to be found in field. For this purpose, extra quantities of control soil were collected, sieved (5 mm mesh) and stored at the lab. Soil samples were treated using the formulated products at nominal concentrations of:

- Avoidance tests with *Eisenia andrei* Bouché, 1972: Roundup® Original (89.29 mg a.e. Kg⁻¹); Trop® (89.29 mg a.e. Kg⁻¹); Zapp® QI 620 (64.29 mg a.e. Kg⁻¹); Crucial® (59.52 mg a.e. Kg⁻¹);
- Reproduction test with *Folsomia candida* Willem, 1902: Roundup® Original (6; 12; 24 e 48 mg a.e. Kg⁻¹); Trop® (6; 12; 24 e 48 mg a.e. Kg⁻¹); Zapp® QI 620 (7.75; 15.5; 31 e 62 mg a.e. Kg⁻¹); Crucial® (8.73; 17.45; 34 e 69.8 mg a.e. Kg⁻¹).

These tests started in the same day of product's addition.

2.4. Test organisms

Earthworms of the species *E. andrei*, collembolans of the species *F. candida* and isopods of the species *Porcellio dilatatus* Brandt, 1833 were obtained from laboratory cultures, maintained under photoperiod of 12 h light: 12 h dark and temperature of 20 ± 1 °C. Earthworms and collembolans were cultured according to ISO guidelines 17512-1 (ISO, 2008) and 11267 (ISO, 2014), respectively. Isopods were cultured following recommendations from the literature (Garcia, 2004; Niemeyer et al., 2009), because no standardized ISO guideline is currently available. However, isopods are considered good candidates for future standardization because of their easily manipulation and role in soil ecosystems (Van Gestel, 2012).

Earthworms were cultured in plastic boxes of 11 L of capacity, in a mixture of cattle dung free of antibiotics and coconut dust (1:1, v:v), moistened with distilled water, fed weekly with cooked oat. Collembolans were cultured in culture vessels lined with a mixture of plaster of Paris and activated charcoal in a ratio of 10:1, fed twice a week with granulated dry yeast *Saccharomyces cerevisiae*. Isopods were cultured in culture vessels lined with worm humus, weekly moistened with sprays of distilled water, fed with slices of potatoes and fish food.

Synchronized organisms were used in the tests: *E. andrei* (2–12 months old; 250–600 mg weight and clitellated), *F. candida* (adults in avoidance tests; 10–12 days old in reproduction tests), and *P. dilatatus* (adult isopods with synchronized weight of 35 ± 10 mg).

2.5. Avoidance behaviour tests

Avoidance tests were carried out following ISO guidelines: ISO 17512-1 (ISO, 2008) for earthworms and ISO 17512-2 (ISO, 2011) for collembolans.

The conventional avoidance tests were carried out with the species *E. andrei* (earthworms) and *F. candida* (collembolans), combining soil from field control soil versus soil from field treatments.

The avoidance test with earthworms was performed using test vessels divided into two equal sections with a removable plastic divider, where one side received control soil (300 g) and the other side the contaminated soil (300 g). A dual control test using control soil in both sides was carried out to check the random distribution of organisms when there is no contamination (ISO, 2008). All the concentrations were tested with five replicates. After putting the soil, the divider was removed and 10 earthworms placed in the centre of the test vessels. The tests were incubated for 48 h at 20 °C $\pm$ 1 with a photoperiod of 12 h:12 h (light:dark), after that, the divider was putted back to separate the recipient again and the number of earthworms in the compartments was counted.

The avoidance test with collembolans was performed similarly to that of earthworms, differentiating in the test vessels content (30 g in each side) and in number of organisms (20 instead of 10). At the end of the test, the number of organisms in each side of the test vessel was recorded using water and stamp ink, following instructions described in the ISO guideline.

2.6. Multispecies avoidance tests with oat straw

These tests were based on ISO 17512-1 (ISO, 2008) and were conducted using *Porcellio dilatatus* (isopods) and *E. andrei* (earthworms). These epigeic species were exposed together, combining oat straw from field control versus oat straw from field treatments, with the aim of checking if these organisms feed on or avoid oat straw with glyphosate application.

For each combination, five boxes received a layer of humus (2 cm) with a thick layer of oat straw overhead (3 cm), moistened with sprays of distilled water. In each replicate, six isopods and ten earthworms were laid in the middle of the box. After 48 h of exposition, a divider was introduced to separate control and contaminated oat straw, and the number of organisms in each side of the box was recorded.

2.7. Reproduction tests with collembolans

These tests were carried out following the ISO 11267 (ISO, 2014) with the collembolans *F. candida*. At the beginning 10 organisms were put into test vessels containing 30 g of contaminated or control soil with approximately 2 mg of dry yeast in each one. Twice a week the recipients were opened allowing aeration and adding water when necessary. After 28 days, the content of each test vessel was carefully transferred into bigger recipients and filled up with water and some drops of stamp ink in order to facilitate the identification of organisms. After careful stirring, adults and juveniles at the surface water were photographed and counted using image analysis software ImageJ (Schneider et al., 2012).

2.8. Presence of earthworms in the field experiment

Two monoliths of 30 $\times$ 30 $\times$ 25 cm were excavated at each field treatment to check the presence of earthworms, following recommendations of Tropical Soil Biology and Fertility (TSBF) methodology (Anderson and Ingram, 1993). This sampling was carried out in the same day that bait lamina test was removed. Only presence or absence of earthworms was recorded.

2.9. Bait lamina test

Bait-lamina is considered as a rapid way to measure site-specific

feeding activity of soil fauna (Römbke et al., 2006). This test was standardized by ISO 18311 (ISO, 2016) and consists of inserting vertically 16-hole-bearing plastic strips filled with 70% cellulose, 25% wheat flour and 5% activated charcoal. The baits were exposed in situ at the 5th day after the herbicides' application, in November 2016, consisting of three groups of eight baits per plot (2 $\times$ 2 m), inserted in three plots per treatment. After 40 days of exposure, the percentage of consumption was determined by recording the number of empty holes (>30% of bait consumed). Time of exposure was extended to 40 days because previous experiences showed the lower consumption of baits in this mesoregion (unpublished data).

2.10. Statistical analysis

Results obtained in avoidance tests were analysed with Fisher's Exact test ($p < 0.05$) allowing comparisons of observed animal's distribution in relation to an expected distribution, assuming the nonexistence of avoidance response (Natal-Da-Luz et al., 2004).

Chronic test's results with *F. candida* determined mean number of juveniles and standard deviation. Results for mean number of juveniles were compared using one-way Analysis of Variance (ANOVA). To confirm normal distribution of data, Shapiro Wilk's test was used and to check up homogeneity of variance the Bartlett Test was used. As a post-hoc test for multiple-comparisons Dunnett's test ($p < 0.05$) was used.

Feeding activity in treatments using bait-lamina was analysed using ANOVA. Normal distribution was checked by Shapiro Wilk's test and Kurtosis, while homogeneity of variance was checked by the Levene's Test. Dunnett's post-hoc test was used for multiple-comparisons among treatments ($p < 0.05$). Data were analysed by generalized linear models (GLM) with the objective to analyse differences between treatments in soil depth.

3. Results and discussion

3.1. Avoidance behaviour tests

After 48 h of exposure, lethality observed for earthworms, collembolans and isopods in avoidance tests remained <10%, and a random distribution was found in dual control tests (control $\times$ control), following the ISO criteria of validation.

On farm soil whereabouts glyphosate was applied, non-significant response could be noticed (Fig. 1a). Result were in accordance with Santos et al. (2012), which found non avoidance of earthworms in field recommended doses of Montana[®] (glyphosate 308 g.L⁻¹ a.i.) in Portugal. However, Casabé et al. (2007) observed avoidance of earthworms in farm soil where Roundup[®] was applied in soybean in Argentina.

The collembolans *F. candida* showed a significant avoidance behaviour ($p < 0.05$) in Zapp[®] Qi 620 treatment (Fig. 1b) while non-avoidance was observed to the other products. It could be motivated by the product formulation with potassium salt. Alves et al. (2015) found effects of vinasses' addition on soil, which was attributed to the vinasses' high salt content and specially the high potassium concentrations. Furthermore, *F. candida* have demonstrated more sensitivity than earthworms in pesticides application (Daam et al., 2011; Santos et al., 2012; Alves et al., 2014), which reinforce the importance of considering these organisms in risk assessment of new products.

Even in avoidance behaviour with *E. andrei* using lab-contaminated soil with high doses of glyphosate, no significant differences ($p > 0.05$) were found except to Crucial[®] (59.52 mg a.e. Kg⁻¹). (Fig. 1e). These results contradict the ones found by Buch et al. (2013) in which the concentration of 30 mg a.i. kg⁻¹ resulted in avoidance, using the formulated product Pica Pau[®] 480 SC

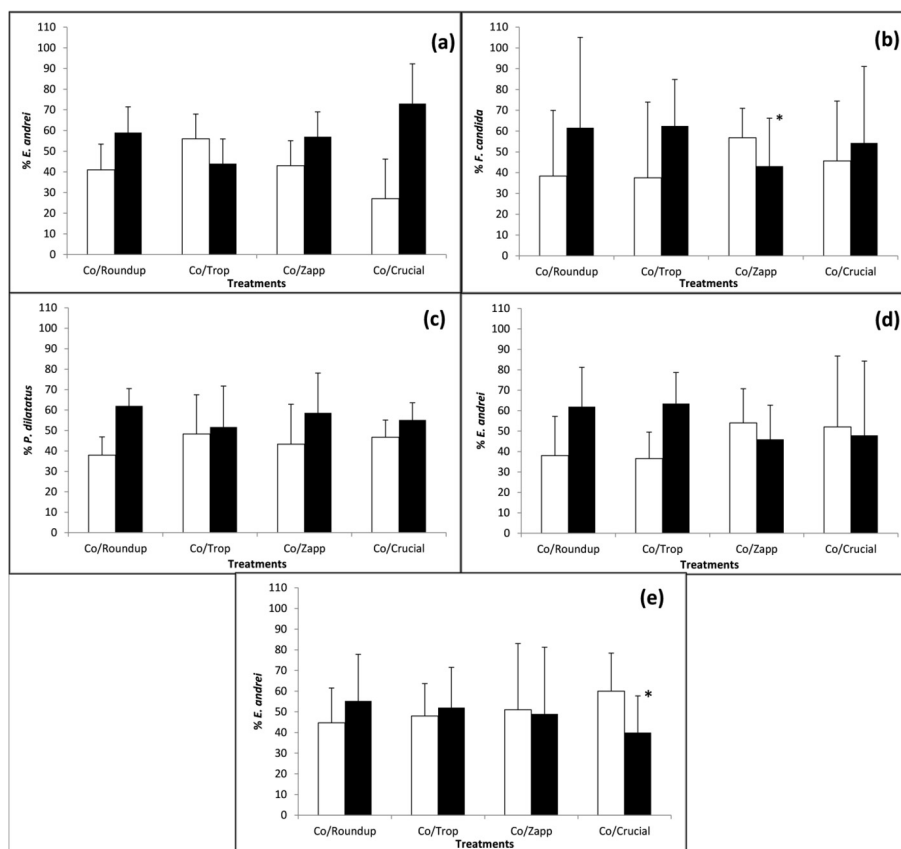


Fig. 1. Avoidance tests: Percentage of organisms on contaminated soil (black bars) against the control (white bars): (a) *E. andrei* and (b) *F. candida* on farm soil in single species tests; (c) *P. dilatatus* and (d) *E. andrei* on oat straw in multispecies test; (e) *E. andrei* on lab-contaminated soil. Asterisks indicate statistical differences ($p < 0.05$).

(glyphosate 480 g.L⁻¹a.i.). The authors obtained the same results for the native earthworm *Pontoscolex corethrurus* in higher doses (47 mg a.i. kg⁻¹) and therefore considered that *E. andrei* acts as a protective indicator for earthworms species.

Otherwise, *E. andrei* seems to be more resistant to pesticides than other earthworm's species (Pelosi et al., 2014). At the present it is not possible to generalize because there is a gap about sensitivity of native species specially in developing countries (Niva et al., 2016; Niemeyer et al., 2017).

3.2. Multispecies avoidance tests with oat straw

In multispecies test, lethality observed for earthworms (*E. andrei*) and isopods (*P. dilatatus*) was <10%, and a random distribution was found in dual control tests (control x control), following the ISO criteria of validation.

Non-avoidance behaviour was observed to oat straw from the treatments (Fig. 1c and d). The results are in accordance to Santos et al. (2010) who found avoidance response of the terrestrial isopod *Porcelionides pruinosus* just in high concentrations (AC₅₀ = 39.7 mg a.i. kg⁻¹) applying Roundup® in LUFA 2.2 soil.

Isopods have already proved high sensitivity to insecticides (Jänsch et al., 2005; Daam and Van Den Brink, 2010) while their usefulness to assess the relative toxicity of the litter layer (Drobne, 1997) could be providential in herbicides field evaluation.

3.3. Reproduction tests with collembolans

After 28 days of exposure, the lethality in controls (<20%), juvenile production per replicate (≥100) and variation coefficient of

reproduction (≤30%) met the requirements for validation in *F. candida* chronic test. No significant difference ($p > 0.05$) between contaminated and control soils in number of juveniles could be observed (Table 1).

In Santos et al. (2010), reproduction of *F. candida* was impaired when exposed to Roundup Original®, showing a low EC₅₀ value

Table 1

Reproduction success of the collembolans *F. candida* exposed to different doses of formulated products in soil contaminated in laboratory (nominal concentrations).

Treatment	mg a.e. kg ⁻¹	<i>F. candida</i> Number of juveniles
Control		312.40 ± (38.97)
Roundup Original®	48	313.25 ± (23.75)
	24	371.80 ± (36.60)
	12	351.40 ± (94.67)
	6	342.40 ± (91.26)
Trop®	48	296.60 ± (36.86)
	24	295.75 ± (79.61)
	12	384.20 ± (33.07)
	6	365.20 ± (67.70)
Zapp® Qi 620	62	351.60 ± (89.74)
	31	445.00 ± (41.85)
	15.5	455.20 ± (95.75)
	7.75	379.60 ± (44.34)
Crucial®	69.8	381.80 ± (73.58)
	34.9	381.20 ± (71.73)
	17.45	368.60 ± (68.73)
	8.725	435.00 ± (120.37)

(0.33 mg a. i. kg⁻¹) in LUFA 2.2 soil, indicating impact on reproductive capacity of this collembolan species. A possible reason for this discrepancy might be related to soil characteristics, which can control glyphosate bioavailability or degradation which is influenced by several factors in interaction (pH, organic matter, biological activity) that can even have opposite effects, for example, soil organic matter can decrease glyphosate bioavailability by adsorption, or its content can improve biological activity and consequently glyphosate degradation in soil (Okada et al., 2017).

Results obtained by Santos et al. (2012) are consistent with the results found by this study. The authors evaluated the impact of the herbicide Montana®, and concluded that it does not seem to affect either earthworms or collembolans at the recommended field dose.

3.4. Presence of earthworms in field experiment

Earthworm's presence in all field plots corroborates the results found in avoidance tests. Although earthworms being considered the main consumers of bait material (Förster et al., 2004; Römbke et al., 2006), the fact that these organisms were found in field suggests that not only macrofauna acts in feeding, but other groups such as meso and microfauna. However, long term effects to earthworms were not considered in this screening.

Available data regarding effects of glyphosate to earthworms are contradictory. While ecotoxicity has been observed only in very high concentrations in laboratory tests (e.g., García-Torres et al., 2014; von Mérey et al., 2016), some field studies have found lower abundance, activity or effects on reproductive capacity of earthworms at recommended doses or lower ones (García-Pérez et al., 2014, 2016; Gaupp-Berghausen et al., 2015; Stellin et al., 2017).

Further steps of this research should encompass chronic effects on earthworm's reproduction and field abundance (specially of native species) to better understand the effects of glyphosate on this group, once earthworms are considered ecosystem engineers in agroecosystems and significantly contribute to ecosystem functioning in these environments.

3.5. Bait-lamina test

After 40 days, the bait-lamina test showed a similar consumption (Fig. 2) in all treatments (Control 63.4 ± 10.7%; Roundup® Original 69.2 ± 25.9%; Trop® 57.7 ± 2.2%; Zapp® Qi 620 68.2 ± 13.8%) except on Crucial®'s plots (37.9 ± 7%). This could be explained by the high toxicity of this product, formulated as glyphosate isopropylamine and potassium salt, classified as "Class I - Red Band - Extremely Toxic" by Brazilian's Ministry of Agriculture, Livestock and Food Supply. Considering the consumption in depth, significant

differences were found at 4.5 and 7 cm (Fig. 3).

The current literature on effects of herbicides to non-target organisms suggests that differences in the ecotoxicity between glyphosate formulations can be attributed to the so-called inert products in the formulation (Pereira et al., 2009; Piola et al., 2013). According to Abraham (2009) the components of the solution that can alter the performance of the glyphosate product are the salt and the adjuvant. Carvalho et al. (2015) evaluating the effect of different formulations of glyphosate in weed species *Commelina benghalensis* verified that the ammonium salt formulation (Roundup® WG) and the formulation of potassium and isopropylamine salt mixture (Crucial®) were the most efficient in controlling this weed. For soil organisms, we can suppose that combination of salts and adjuvants present in the Crucial® formulation may have contributed to the lower consumption of the baits. Another hypothesis is that the association of the isopropylamine and potassium salts may have resulted in a synergistic effect making this formulation more toxic than the others. Contardo-Jara et al. (2009) showed that the active ingredient toxicity for Roundup® presented higher toxicity than glyphosate itself probably due to the surfactant. This shows the need for adequacy in risk assessment rules to require evaluation of the agrochemical formulated product and not only the active ingredient.

Casabé et al. (2007) in Argentina found significantly lower consumption of bait lamina after 50 days of exposure in soybean with application of Roundup® (15.5 ± 4.4%) in comparison to the control without herbicide (24.0 ± 7.0%), which was attributed to low density of macrofauna or event soil compaction. Otherwise, Römbke et al. (2006) measuring the feeding activity in Amazonian forest ecosystems hypothesize that oribatid mites could be the mainly responsible for the failure of correlation found in those sites, leading to conclude that mesofauna have an important role in top surface processes. In the present study, the fact that earthworms were found in all treatments may be an indicative that not only macrofauna, but other groups like microorganisms and mesofauna can act as consumers of bait lamina, and that they could be impacted in application of Crucial®.

Other possibility is a disruption in ecological interactions by herbicide application, which is still poorly understood. Benamú et al. (2010) found sublethal effects of glyphosate on arachnid's reproduction, web building and prey consumption. These effects on predators (as arachnids) could reflect on differences in community composition in macro and mesofauna community where the herbicide is applied, and consequently change the pattern of organic material consumption. Results of Zaller et al. (2014) showed that Roundup can affect interactions between earthworms and arbuscular mycorrhizal fungi, which can led to consequences for crop cultivation.

Furthermore, the results of bait lamina suggest that the avoidance behaviour tests with earthworms, isopods and collembolans, and the reproduction test with collembolans, used as screening in the present work, were not sufficient to predict functional impacts in the field.

Other interesting question to be further understood is how long the impairment on feeding activity can last, and if this impact has consequences to plant productivity. Despite the considered short half-life of glyphosate be less than 25–47 d in soil, (Szekacs and Darvas, 2013; Myers et al., 2016), its metabolic AMPA is more persistent than glyphosate, remaining in soil between 119 and 958 days (Cox, 1998).

4. Conclusions

Recommended doses of the tested products, in one application, did not cause avoidance behaviour of earthworms in soil, or

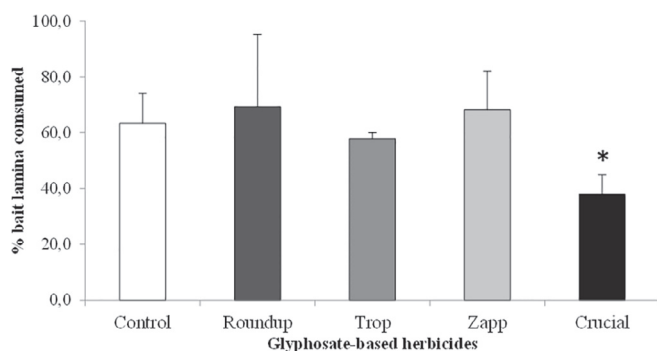


Fig. 2. Bait-lamina test. Percentage of food activity after 40 days of exposure of the baits. Asterisks indicates statistical difference ($p < 0.05$).

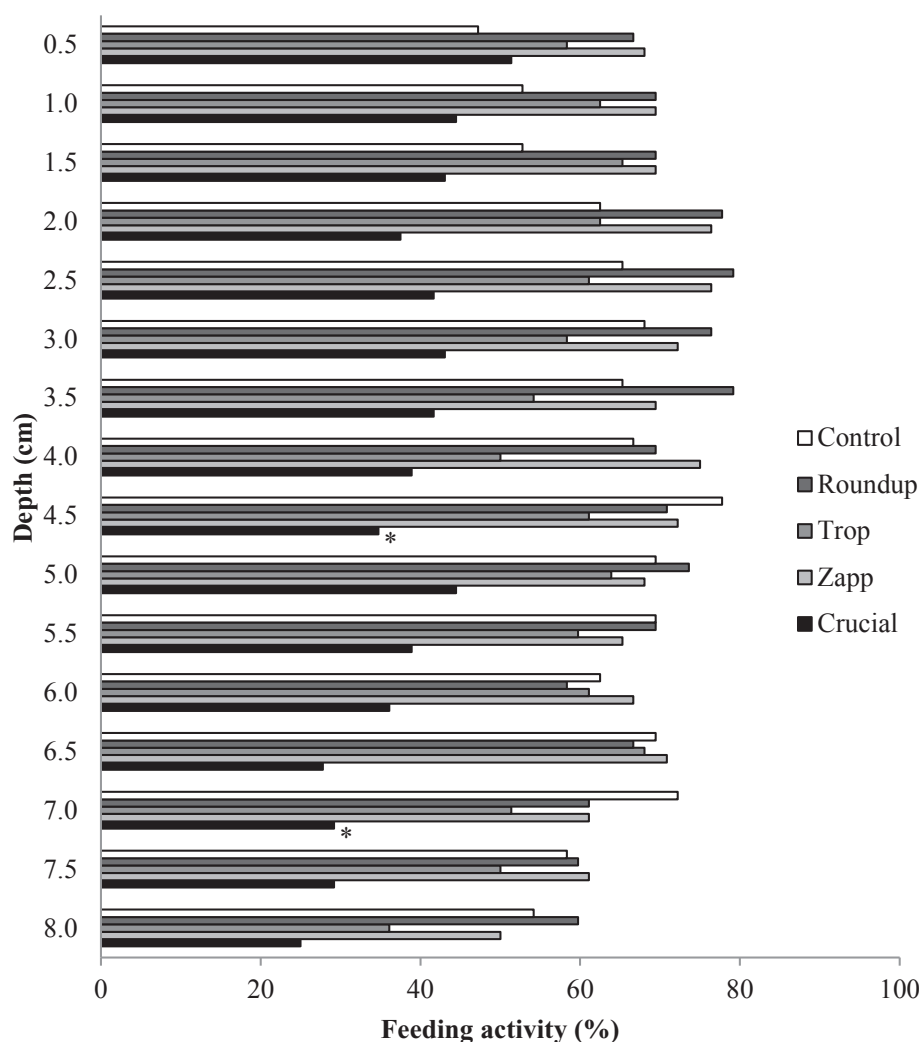


Fig. 3. Bait-lamina test. Percentage consumption of baits after 40 days of exposure in field after application of commercial herbicides, according to depth. Asterisks indicate statistical difference ($p < 0.05$) among treatments and control in each depth.

avoidance of earthworms and isopods exposed to oat straw. Collembolans avoided the field soil with Crucial® even at the recommended dose. The presence of earthworms was confirmed by field sampling.

Laboratory spiked samples did not caused avoidance of earthworms even at high concentrations of glyphosate (10× higher than the expected to reach the soil after application). Once these organisms did not avoid these products, further evaluations should be evolved to assess chronic effects to their populations.

Field evaluation with bait lamina indicated impairment of feeding activity of soil fauna in Crucial® treatment. Among the tested products, it was the only classified as extremely toxic (red label). Such impairment was not predicted by avoidance behaviour tests, earthworms presence in field, neither by collembolan reproduction tests. It can be related to chronic effects to soil fauna or impacts on other groups or species not evaluated in this study, e.g., acari, enchytraeids or even microorganisms. The results suggest that earthworm's endpoints could not be sufficient to fully protect other organism or processes.

The results showed the relevance of bait lamina test on screening herbicides' impact in the field. Further research should verify if such effects pointed by bait lamina have some impact in crop production, and how long such impairment in feeding activity

can last.

This work reinforces the importance of considering different formulations for the same active ingredient in risk assessment of pesticides.

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References

- Abraham, W., 2009. Formulações de glyphosate e adjuvantes. In: Velini, E.D., Meschede, D.K., Carbonari, C.A., Trindade, M.L.B. (Eds.), *Glyphosate. Feap, Botucatu*, pp. 179–190.
- Aguilar, L.M., De Figueira, F.H., Gottschalk, M.S., Da Rosa, C.E., 2016. Glyphosate-based herbicide exposure causes antioxidant defence responses in the fruit fly *Drosophila melanogaster*. *Comp. Biochem. Physiol. C Toxicol. Pharmacol.* 185–186, 94–101.
- Allegrini, M., Zabaloy, M.C., Gomez, E., Del, V., 2015. Ecotoxicological assessment of soil microbial communities tolerance to glyphosate. *Sci. Total Environ.* 533, 60–68.
- Alves, P.R.L., Cardoso, E.J.B.N., Martinez, A.M., José PauloSousa, J.P., Pasini, A., 2014. Seed dressing pesticides on springtails in two ecotoxicological laboratory tests.

- Ecotoxicol. Environ. Saf. 105, 65–71.
- Alves, P.R., Natal-da-Luz, T., Sousa, J.P., Cardoso, E.J., 2015. Ecotoxicological characterization of sugarcane vinasses when applied to tropical soils. *Sci. Total Environ.* 526, 222–232.
- Anderson, J.M., Ingram, J.S.I., 1993. *Tropical Soil Biology and Fertility: a Handbook of Methods*, 2 ed. CAB International, Wallington, UK, p. 256.
- Benamí, M.A., Schneider, M.L., Sánchez, N.E., 2010. Effects of the herbicide glyphosate on biological attributes of *Alpaida veniliae* (Araneae, Araneidae), in laboratory. *Chemosphere* 78, 871–876.
- Benbrook, C.M., 2016. Trends in glyphosate herbicide use in the United States and globally. *Environ. Sci. Eur.* 28 (3).
- Buch, A.C., Brown, G.G., Niva, C.C., Sautter, K.D., Sousa, J.P., 2013. Toxicity of three pesticides commonly used in Brazil to *Pontosclex corethrurus* (Muller, 1857) and *Eisenia andrei* (Bouché 1972). *Appl. Soil Ecol.* 69, 32–38.
- Carneiro (Org.), F.F., 2015. Dossiê ABRASCO: um alerta sobre os impactos dos agrotóxicos na saúde/Organização de Fernando Ferreira Carneiro, Lia Giraldo da Silva Augusto, Raquel Maria Rigotto, Karen Friedrich e André Campos Búrgio. EPSJV, Rio de Janeiro. São Paulo: Expressão Popular. 624 pp.
- Carvalho, F.P., José, R.H.S., Lopes, L.C.M., Ronchi, C., 2015. Benghal dayflower control with diferente glyphosate formulations. *Rev. Bras. Hist.* 14, 194–199.
- Casabé, N., Piola, L., Fuchs, J., Oneto, M.L., Pamparato, L., Basack, S., Giménez, R., Massaro, R., Papa, J., Kesten, E., 2007. Ecotoxicological assessment of the effects of glyphosate and chlorpyrifos in an Argentine soya field. *J. Soils Sediments* 7, 232–239.
- Contardo-Jara, V., Klingelmann, E., Wiegand, C., 2009. Bioaccumulation of glyphosate and its formulation Roundup Ultra in *Lumbriculus variegatus* and its effects on biotransformation and antioxidant enzymes. *Environ. Pollut.* 157, 57–63.
- Cox, C., 1998. Herbicide factsheet: glyphosate (Roundup). *J. Pestic. Reform* 18, 1–15.
- Daam, M.A., Van Den Brink, P.J., 2010. Implications of differences between temperate and tropical freshwater ecosystems for the ecological risk assessment of pesticides. *Ecotoxicology* 19, 24–37.
- Daam, M.A., Leitão, S., Cerejeira, M.J., Sousa, J.P., 2011. Comparing the sensitivity of soil invertebrates to pesticides with that of *Eisenia fetida*. *Chemosphere* 85, 1040–1047.
- Dominguez, A., Brown, G.G., Sautter, K.D., Oliveira, C.M.R., Vasconcelos, E.C., Niva, C.C., Bartz, M.L.C., Bedano, J.C., 2016. Toxicity of AMPA to the earthworm *Eisenia andrei* Bouché, 1972 in tropical artificial soil. *Sci. Rep.* 6, 1973.
- Drobne, D., 1997. Terrestrial isopods – a good choice for toxicity testing of pollutants in the terrestrial environment. *Environ. Toxicol. Chem.* 16, 1159–1164.
- Förster, B., Van Gestel, C.A.M., Koolhaas, J.E., Nentwig, G., Rodrigues, J.M.L., Sousa, J.P., Jones, S.E., Knacker, T., 2004. Ring-Testing and field-validation of a terrestrial model ecosystem (TME)—an instrument for testing potentially harmful substances: effects of carbendazim on organic matter breakdown and soil fauna feeding activity. *Ecotoxicology* 13, 129–141.
- Garcia, M., 2004. Effects of pesticides on soil fauna: Development of ecotoxicological test methods for tropical regions. *Ecol. Dev. Ser.* 19, 291.
- García-Pérez, J.A., Alarcón-Gutiérrez, E., Perroni, Y., Barois, I., 2014. Earthworm communities and soil properties in shaded coffee plantations with and without application of glyphosate. *Appl. Soil Ecol.* 83, 230–237.
- García-Pérez, J.A., Alarcón-Gutiérrez, E., Hernandez, Y., Hernandez, C., 2016. Impact of litter contaminated with glyphosate-based herbicide on the performance of *Pontosclex corethrurus*, soil phosphatase activities and soil pH. *Appl. Soil Ecol.* 104, 31–34.
- García-Torres, T., Giuffré, L., Romaniuk, R., Ríos, R.P., Pagano, E.A., 2014. Exposure assessment to glyphosate of two species of annelids. *Bull. Environ. Contam. Toxicol.* 93, 209–214.
- Gaupp-Berghausen, M., Hofer, M., Rewald, B., Zaller, J.G., 2015. Glyphosate-based herbicides reduce the activity and reproduction of earthworms and lead to increased soil nutrient concentrations. *Sci. Rep.* 5, 12886.
- Giesy, J.P., Dobson, S., Solomon, K.R., 2000. Ecotoxicological risk assessment for Roundup® herbicide. *Rev. Environ. Contam. Toxicol.* 167, 35–120.
- ISO - International Organization For Standardization, 2008. ISO 17512–1: Soil Quality – Avoidance Test for Determining the Quality of Soils and Effects on Behaviour – Part 1: Test with Earthworms (*Eisenia fetida* and *Eisenia andrei*). Geneva.
- ISO, 2011. ISO 17512–2: Soil Quality - Avoidance Test for Determining the Quality of Soils and Effects of Chemicals on Behaviour - Part 2: Test with Collembolans (*Folsomia candida*). Geneva.
- ISO, 2014. ISO 11267: Soil Quality – Inhibition of Reproduction of Collembola (*Folsomia candida*) by Soil Contaminants. Geneva.
- ISO, 2016. ISO 18311: Soil Quality - Method for Testing Effects of Soil Contaminants on the Feeding Activity of Soil Dwelling Organisms - Bait-lamina Test. Geneva.
- Jänsch, S., Garcia, M., Römbke, J., 2005. Acute and chronic isopod testing using tropical *Porcellionides pruinosus* and three model pesticides. *Eur. J. Soil Biol.* 41, 143–152.
- Kanashiro, M.M., 2015. Avaliação da toxicidade do glifosato e da deltametrina em solo. Trabalho de Conclusão de Curso. Escola de Engenharia de São Carlos. Universidade de São Paulo, São Carlos.
- Lane, M., Lorenz, N., Saxena, J., Ramsier, C., Dick, R.P., 2012. The effect of glyphosate on soil microbial activity, microbial community structure, and soil potassium. *Pedobiologia* 55, 335–342.
- Maqueda, C., Undabeytia, T., Villaverde, J., Morillo, E., 2017. Behaviour of glyphosate in a reservoir and the surrounding agricultural soils. *Sci. Total Environ.* 593–594, 787–795.
- Myers, J.P., Antoniou, M.N., Blumberg, B., Carroll, L., Colborn, T., Everett, L.G., Hansen, M., Landrigan, P.J., Lanphear, B.P., Mesnage, R., Vandenberg, L.N., Vom Saal, F.S., Welshons, W.V., Benbrook, C.M., 2016. Concerns over use of glyphosate-based herbicides and risks associated with exposures: a consensus statement. *Environ. Health* 17, 15–19.
- Natal-da-Luz, T., Ribeiro, R., Sousa, J.P., 2004. Avoidance tests with collembola and earthworms as early screening tools for site specific assessment of polluted soils. *Environ. Toxicol. Chem.* 24, 2188–2193.
- Niemeyer, J.C., Santos, V.C., Araújo, P.B., Da Silva, E.M., 2009. Reproduction of *Cubaris murina* (Crustacea: isopoda) under laboratory conditions and its use in ecotoxicity tests. *Braz. J. Biol.* 69, 137–142.
- Niemeyer, J.C., Chelinho, S.C., Sousa, J.P., 2017. Soil ecotoxicology in Latin America: current research and perspectives. *Environ. Toxicol. Chem.* 36, 1795–1810.
- Niva, C.C., Niemeyer, J.C., Júnior, F.M., Nunes, M.E., De Sousa, D.L., Aragão, C.W., Sautter, K.D., Espindola, E.G., Sousa, J.P., Römbke, J., 2016. Soil ecotoxicology in Brazil is taking its course. *Environ. Sci. Pollut. Res. Int.* 23, 11363–11378.
- Okada, E., Costa, J.L., Bedmar, F., 2017. Glyphosate dissipation in different soils under no-till and conventional tillage. *Pedosphere* (in press).
- Pelosi, C., Barot, S., Capowiez, Y., Hedde, M., Van Den Bulcke, F., 2014. Pesticides and earthworms. A review. *Agron. Sustain. Dev.* 34, 199–228.
- Pereira, J.L., Antunes, S.C., Castro, B.B., Marques, C.R., Gonçalves, A.M., Gonçalves, F., Pereira, R., 2009. Toxicity evaluation of three pesticides on non-target aquatic and soil organisms: commercial formulation versus active ingredient. *Ecotoxicology* 18, 455–463.
- Piola, L., Fuchs, J., Oneto, M.L., Basack, S., Kesten, E., Casabé, N., 2013. Comparative toxicity of two glyphosate-based formulations to *Eisenia andrei* under laboratory conditions. *Chemosphere* 91, 545–551.
- Römbke, J., Hofer, H., Garcia, M.V.B., Martius, C., 2006. Feeding activities of soil organisms at four different forest sites in Central Amazonia using the bait lamina method. *J. Trop. Ecol.* 22, 313–320.
- Santos, M.J.G., Soares, A.M.V.M., Loureiro, S., 2010. Joint effects of three plant protection products to the terrestrial isopod *Porcellionides pruinosus* and the collembolan *Folsomia candida*. *Chemosphere* 80, 1021–1030.
- Santos, M.J., Ferreira, M.F., Cachada, A., Duarte, A.C., Sousa, J.P., 2012. Pesticide application to agricultural fields: effects on the reproduction and avoidance behaviour of *Folsomia candida* and *Eisenia andrei*. *Ecotoxicology* 21, 2113–2122.
- Schneider, C.A., Rasband, W.S., Eliceiri, K.W., 2012. NIH Image to ImageJ: 25 years of image analysis. *Nat. Med.* 9, 671–675.
- Stellin, F., Gavinelli, F., Stevanato, P., Concheri, G., Squartini, A., Paoletti, M.G., 2017. Effects of different concentrations of glyphosate (Roundup 360®) on earthworms (*Octodrilus complanatus*, *Lumbricus terrestris* and *Aporrectodea caliginosa*) in vineyards in the North-East of Italy. *Appl. Soil Ecol.* (in press).
- Szekacs, A., Darvas, B., 2013. Forty years with glyphosate. In: Hasaneen, M.N.A.E.G. (Ed.), *Herbicides - Properties, Synthesis and Control of Weeds*. InTech, pp. 247–284.
- Tavella, L.B., Silva, I.N., Fontes, L.O., Dias, J.R.M., Silva, M.I.L., 2011. O uso de agrotóxicos na agricultura e suas consequências toxicológicas e ambientais. *ACSA-Agropecuária Científica no Semi-Árido* 7, 6–12.
- USEPA, 2012. Edition of the Drinking Water Standards and Health Advisories. EPA, pp. 2–6.
- Van Gestel, C.A.M., 2012. Soil ecotoxicology: state of the art and future directions. *ZooKeys* 176, 275–296.
- von Mérey, G., Manson, P.S., Mehrsheikh, A., Sutton, P., Levine, S.L., 2016. Glyphosate and aminomethylphosphonic acid chronic risk assessment for soil biota. *Environ. Toxicol. Chem.* 35, 2742–2752.
- Zaller, J.G., Heigl, F., Ruess, L., Grabmaier, A., 2014. Glyphosate herbicide affects belowground interactions between earthworms and symbiotic mycorrhizal fungi in a model ecosystem. *Sci. Rep.* 4, 5634.